

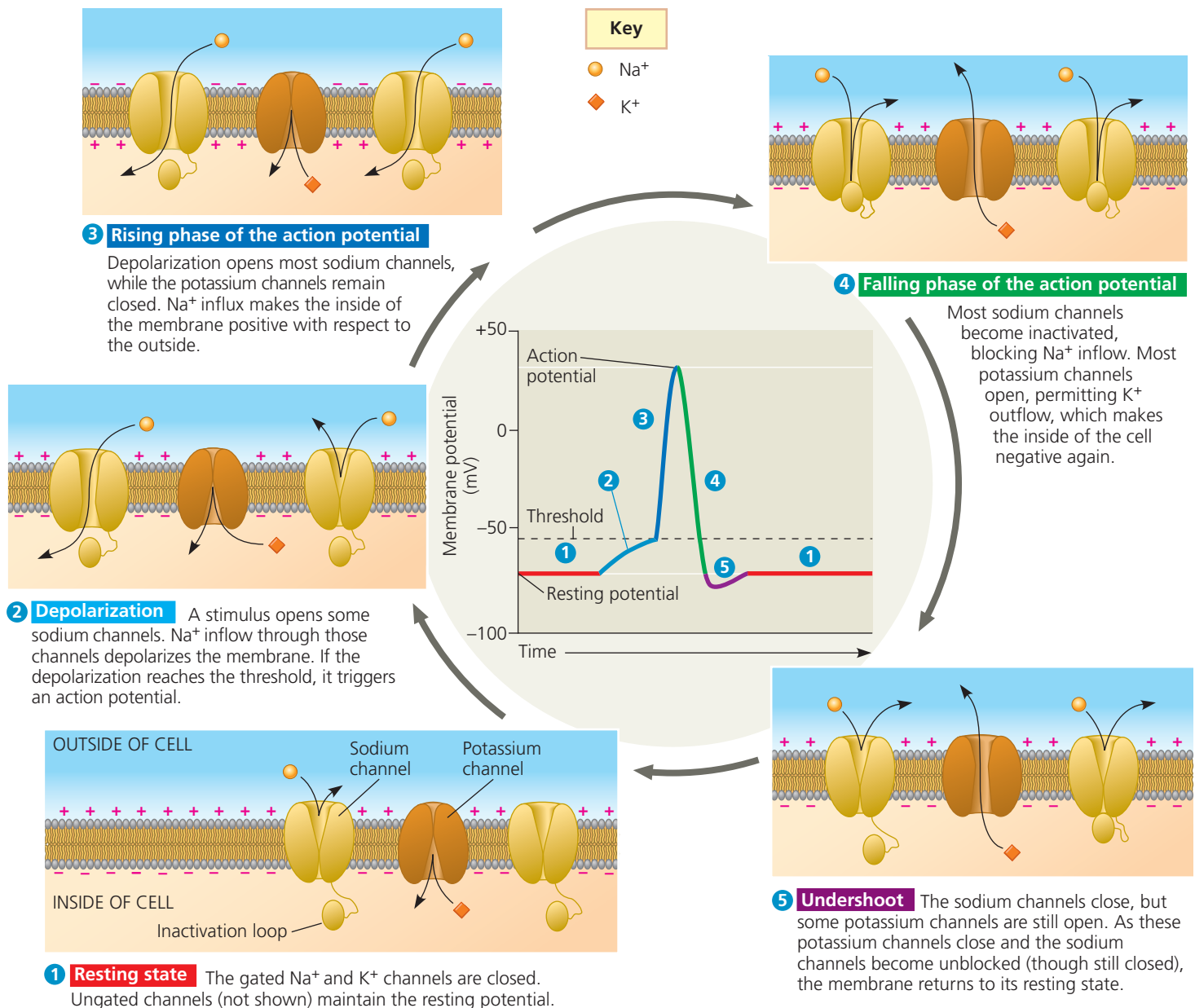
but become *inactivated*: A portion of the channel protein called an inactivation loop blocks ion flow through the open channel. Sodium channels remain inactivated until after the membrane returns to the resting potential and the channels close. Potassium channels open more slowly than sodium channels, but remain open and functional until the end of the action potential.

To understand further how voltage-gated channels shape the action potential, consider the process as a series of stages,

as depicted in **Figure 48.12**. **1** When the membrane of the axon is at the resting potential, most voltage-gated sodium channels are closed. Some potassium channels are open, but most voltage-gated potassium channels are closed. **2** When a stimulus depolarizes the membrane, some gated sodium channels open, allowing more Na^+ to diffuse into the cell. If the stimulus is sufficiently strong, the Na^+ inflow persists, causing further depolarization, which opens more gated sodium channels, allowing even more Na^+ to diffuse into the

▼ Figure 48.12 The role of voltage-gated ion channels in the generation of an action potential.

The circled numbers on the graph in the center and the colors of the action potential phases correspond to the five diagrams showing voltage-gated sodium and potassium channels in a neuron's plasma membrane.



DRAW IT Draw a simple diagram showing how positive feedback underlies the rising phase of the action potential. Your diagram should include these three events: (1) a change in membrane potential; (2) ion flow; and (3) opening, closing, or inactivation of a channel.

➔ **Mastering Biology BioFlix® Animation: Action Potential**